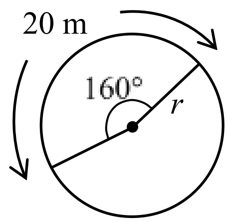
Question 1 2 marks 101

Determine the length of the radius, *r*, given an arc length of 20 metres and a central angle of 160°.



Question 2 — 1 mark 102

There are eight cars parked in a row. Determine the number of possible arrangements of these eight cars if Mrs. Jones must always park in the third spot and Mr. Rodriguez must always park in the last spot.

Question 3 3 marks 103

Bill wins \$1 300 000 in a lottery and invests the entire amount at an annual interest rate of 2.5% compounded quarterly. He will withdraw \$10 000 at the end of every three months.

Determine, algebraically, the total number of withdrawals, including the partial amount, that Bill can make until there is no money left. Express your answer as a whole number.

Use the formula:

$$PV = \frac{R\left[1-(1+i)^{-n}\right]}{i}$$

where PV = the present value deposited

R = the amount of each withdrawal

n = the number of equal withdrawals

$$i = \frac{\text{the annual interest rate (in decimal form)}}{\text{the number of compounding periods}}$$

Question 4 3 marks 104

Determine and simplify the 12th term in the binomial expansion of $\left(x^3 - \frac{1}{2x^2}\right)^{12}$.

Question 5 | 3 marks 105

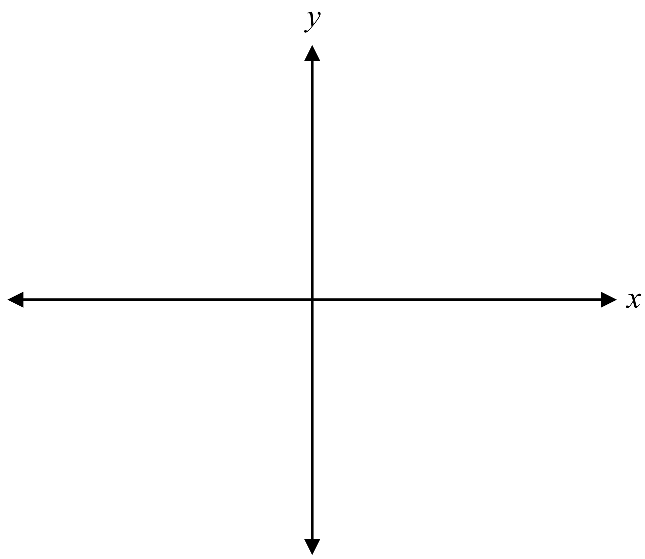
Solve, algebraically.

$$e^{2x-3} = 7^{x+1}$$

Note: A calculator is not required for the remaining test questions.

Question 6 1 mark 106

Sketch the angle $\frac{7\pi}{3}$ in standard position.



Question 7 3 marks 107

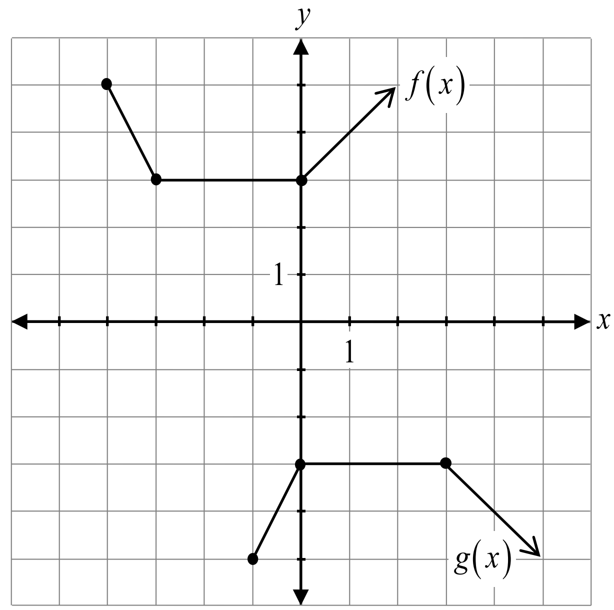
Determine, algebraically, all of the zeros of the polynomial function $P(x) = x^4 - 5x^3 - 4x^2 + 20x$.

Question 8 1 mark 108

Justify why four of the terms in the binomial expansion of $(-x + y)^6$ are positive.

Question 9 2 marks

Determine the equation of $g(x)$ in terms of $f(x)$.



$g(x) =$ ______________________________

Question 10 3 marks 110

Prove the identity for all permissible values of x.

$$\frac{\sin x + \tan x}{\cot x + \csc x} = \frac{\sin^2 x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Question 11 1 mark 111

Given that the point $(-2,1)$ is on the graph of $y = f(x)$, describe how the coordinates of the corresponding point on the graph of $y = f(4x)$ are different.

Question 12 3 marks 112

Using the laws of logarithms, completely expand the expression:

$$\log\left(\frac{x^2\sqrt{y}}{w-1}\right)$$

Question 13 1 mark 113

Given $\sec\theta = -\frac{5}{4}$ and $\tan\theta > 0,$ state the quadrant in which θ terminates.

Justify your answer.

Question 14 — 2 marks 114

State an equation of a rational function that has a vertical asymptote at $x = -8$ and a horizontal asymptote at $y = 9$.

Question 15 — 1 mark 115

Given the point $(5, 1)$, state the coordinates of the corresponding point after a reflection across the line $y = x$.

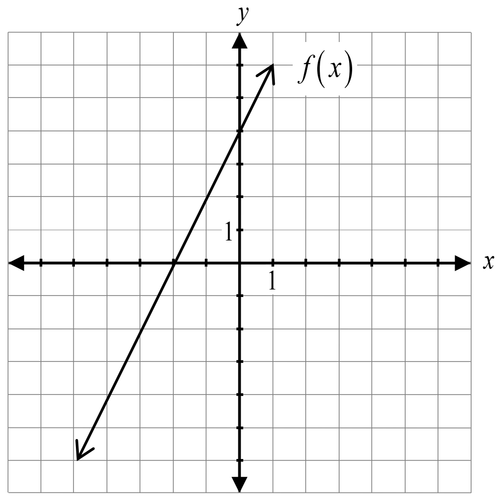
Question 16 1 mark 116

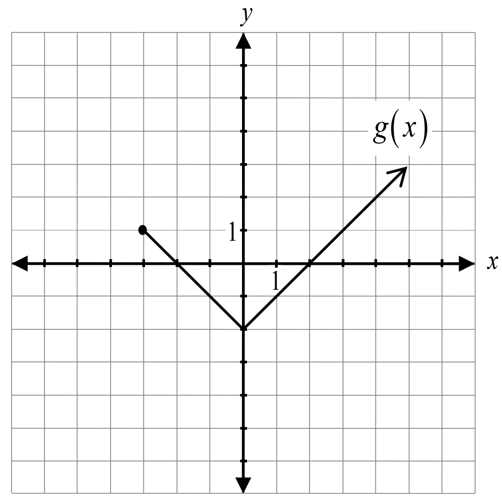
Simplify.

$$\frac{(n-13)!}{(n-12)!}$$

Question 17 a) 1 mark b) 1 mark 117 118

Given the graphs of $y = f(x)$ and $y = g(x)$,





a) determine the value of $f(g(2))$.

b) determine the value of $(g - f)(-3)$.

No marks will be awarded for work done on this page.